

Problem 1 part 1)

Given the image region as shown in Figure 1 and $V = \{1\}$, what is the shortest m-path between p (the pixel at the upper-left corner) and q (the pixel at the bottom-right corner)? Give the length of the path if the path exists. (15 points)

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1 0 1 1 0
0 1 0 1 0
0 0 1 1 0
1 1 0 1 0
0 1 1 1 1

```

Step 1: Identify pixels in V

p

1	0	1	1	0
0	1	0	1	0
0	0	1	1	0
1	1	0	1	0
0	1	1	1	1

q

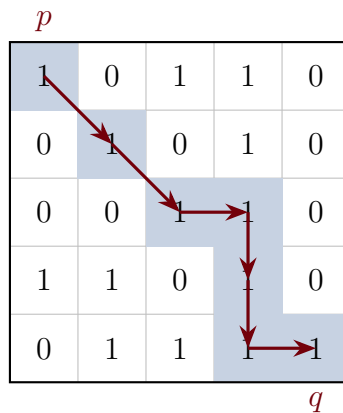
All pixels with value 1 are highlighted. These are the only candidates for the m-path since $V = \{1\}$.

Step 2: Apply m-adjacency rules

Two pixels with values in V are **m-adjacent** if:

- They are 4-adjacent (share an edge), OR
- They are diagonally adjacent AND their shared 4-neighbors contain no pixels with values in V .

The second condition prevents diagonal shortcuts when a 4-connected bridge exists, eliminating path ambiguity.

Step 3: Trace the shortest m-path

Path: $(0, 0) \rightarrow (1, 1) \rightarrow (2, 2) \rightarrow (2, 3) \rightarrow (3, 3) \rightarrow (4, 3) \rightarrow (4, 4)$

Diagonal move justifications:

- $(0, 0) \rightarrow (1, 1)$: Shared 4-neighbors are $(0, 1) = 0$ and $(1, 0) = 0$. Neither is in V , so m-adjacent.
- $(1, 1) \rightarrow (2, 2)$: Shared 4-neighbors are $(1, 2) = 0$ and $(2, 1) = 0$. Neither is in V , so m-adjacent.

Results

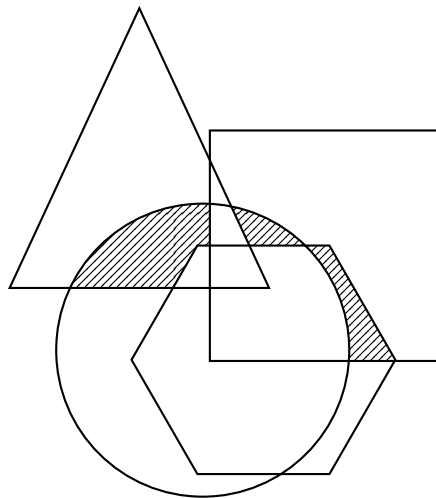
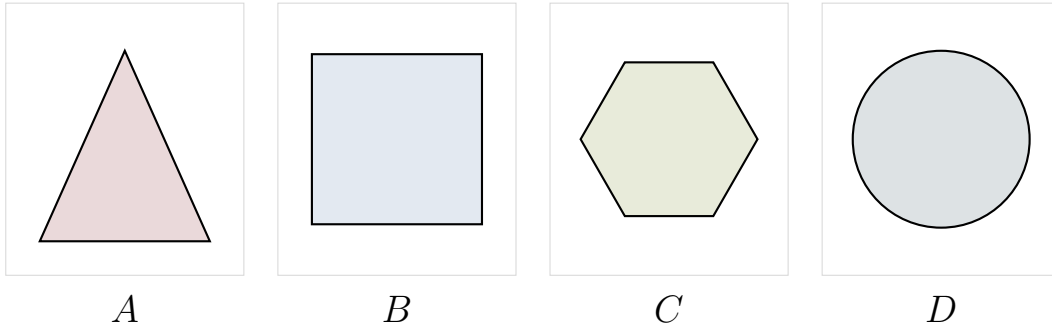
The shortest m-path exists and has **length 6**.

Problem 1 part 2)

To take a picture for fireworks at night, give at least two operations that should be applied to improve the visualization quality of the image and explain. (15 points)

Problem 2)

A, B, C and D are four sets. Give expressions for the set shown shaded in Figure 2 in terms of A, B, C, and D. Note that all of the shaded areas constitute of one set



Problem 3)

Given a 3x3 spatial filter $\frac{1}{9} \begin{bmatrix} -1 & -1 & -1 \\ -1 & -17 & -1 \\ -1 & -1 & -1 \end{bmatrix}$ answer the following questions:

a) What type of filter is it? When applying this filter on an image, what kind of effect you will expect in the output image? Why? (Hint: is this filter an image smoothing filter, an image sharpening filter or an edge detector?) (10 points)

b) Perform the image convolution using this filter on a 3x3 image $\begin{bmatrix} 3 & 5 & 12 \\ 7 & 18 & 11 \\ 5 & 9 & 15 \end{bmatrix}$. Give the CROPPED filtering result for the output image. (Hint: you need to assume an appropriate zero mapping) (15 points)

Problem 4)

An image with intensities in the range $r \in [0, L - 1]$ has the pdf

$$p_r(r) = \begin{cases} \frac{4r}{3(L-1)^2} & 0 \leq r \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

where $L - 1$ is the maximum intensity value. It is desired to transform the intensity levels of this image so that they will have the specified pdf

$$p_z(z) = \begin{cases} \frac{2z}{(L-1)^2} & 0 \leq z \leq L-1 \\ 0 & \text{otherwise} \end{cases}$$

Assume that the intensity value is continuous and find the transformation function $z = T(r)$ in terms of r and z that will accomplish this. (25 points)

(Hints: The transformation function of histogram equalization is given as

$$s = T(r) = (L-1) \int_0^r p_r(w) dw$$

)

Bonus Question

Perform the Fourier transform to a function as shown in Figure 3. (20 points)

(Hint: the definition of the Fourier transform of a function is $F(u) = \int_{-\infty}^{\infty} f(t)e^{-j2\pi ut} dt$)

